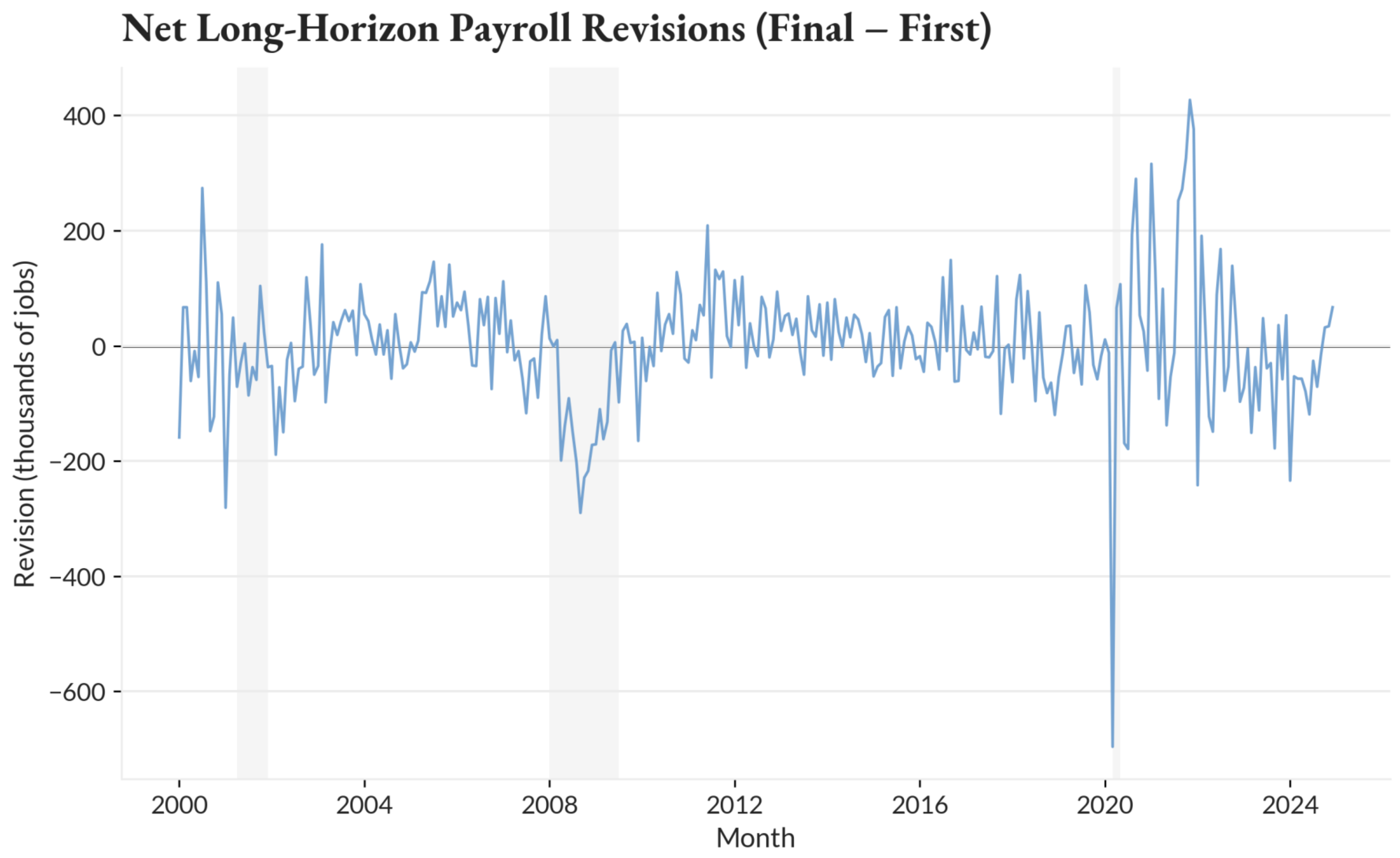
Real-Time Reliability of First-Release Nonfarm Payroll Estimates

Short- and Long-Horizon Evidence from CES Revisions, 2000–2024

Has Payroll Reliability Changed?



Insights

The Reliability Question

- Most revisions are modest, but tails are fat
- Two visible volatility clusters: 2008–09 (GFC) and 2020–21 (COVID era)

Why It Matters

- Policymakers and markets act on first-release data
- Reliability is often treated as stable over time

Core Questions

- Has the revision distribution structurally shifted?
- Or do spikes reflect temporary shocks?

What This Research Does Differently

Conventional Wisdom

The Conventional View

- Revisions are "classical measurement error"—random noise that cancels out
- Reliability assumed constant across time and economic conditions
- Policy models use fixed confidence intervals

The Gap

- No one has quantified when reliability breaks down
- No evidence on how long elevated uncertainty persists
- Decision-making frameworks assume stable reliability

This Paper's Contribution

The Research Framework

- Treat revision variance as regime-dependent, not constant
- Use Interrupted Time Series (COVID as natural experiment)
- Quantify persistence of elevated tail risk

Key Finding

- March 2020 marks a statistically significant level shift
- Tail risk remains elevated 57 months post-shock (nearly 5 years)
- Reliability is dynamic—requires regime-specific confidence intervals

Prior work emphasized point prediction. This research measures distributional uncertainty and regime shifts.

What This Research Does Differently

News vs. Noise
Mankiw & Shapiro (1986)
Aruoba (2008)

Initial releases are evaluated as forecasts of final values, with revisions treated as information updates.

- Debate: Are revisions "news" or "noise"?
- Focus on revision predictability

› **Shifts Focus to Uncertainty**

Real-Time Data
Orphanides (2001)
Croushore & Stark (2001)

Policy conclusions can differ materially when evaluated using real-time rather than revised data.

- Built vintage-dated datasets
- Showed policy errors tied to real-time mismeasurement

› **Quantifies Regime-Dependence**

Labor Measurement
Cajner et al. (2019)
Fed Philadelphia (2023)

Official payroll data can lag labor market conditions, especially around turning points.

- Turning point bias documented
- Early benchmarks have predictive power

› **Measures Persistence**

01

Measurement Framework

Institutional Structure and Empirical Design

How Payroll Data Is Produced—And Revised

Data Sources

CES Survey (First Print)

- Released first Friday of each month by Bureau of Labor Statistics
- Surveys approximately 119,000 business establishments nationwide
- Uses birth-death model to adjust for unobserved firm entry and exit

QCEW Benchmark (Final Count)

- Published with 5-month lag, derived from UI tax records
- Administrative census of 11+ million establishments nationwide
- Achieves >95% employment coverage versus ~33% for CES survey sample

When Reliability Shifts

Normal Times vs. Crisis Periods

- Birth-death model performs well during stable economic expansions
- Turning points invalidate backward-looking statistical adjustments
- Response rates collapse during crises, amplifying sampling variance

The Uncertainty Cascade

- Uncertainty compounds as delayed survey responses arrive sporadically
- Revisions show directional bias during recessions, not random error
- Extreme revisions reveal underlying economic regime transitions

Key Measures

Long-Horizon Revision

- Defined as |Latest-Available − First Print| in absolute magnitude terms
- Captures total uncertainty resolved after administrative data arrives
- Serves as proxy for reliability of real-time labor market signals

Sample Coverage

- 300 monthly observations from January 2000 to December 2024
- Excludes 2025 series due to pending QCEW benchmark revision
- Spans three NBER recessions (2001, 2007-09, 2020) and recovery phases

Three-Stage Empirical Design

Distributional patterns reveal heavy tails and regime-dependent variance.

- Unconditional distributions show median revision near zero but wide dispersion, with upper-tail quantiles rising sharply
- Regime stratification demonstrates exceedance probabilities triple during COVID versus pre-2020 baseline periods

Interrupted Time Series isolates COVID-19 as exogenous shock to reliability.

- ITS design compares pre-COVID baseline (2000-2019) to post-COVID period, testing for structural break in tail risk
- Robustness checks include placebo tests, pre-trend verification, negative controls, and event-study specifications

Predictive models test whether revision risk can be forecast in real time.

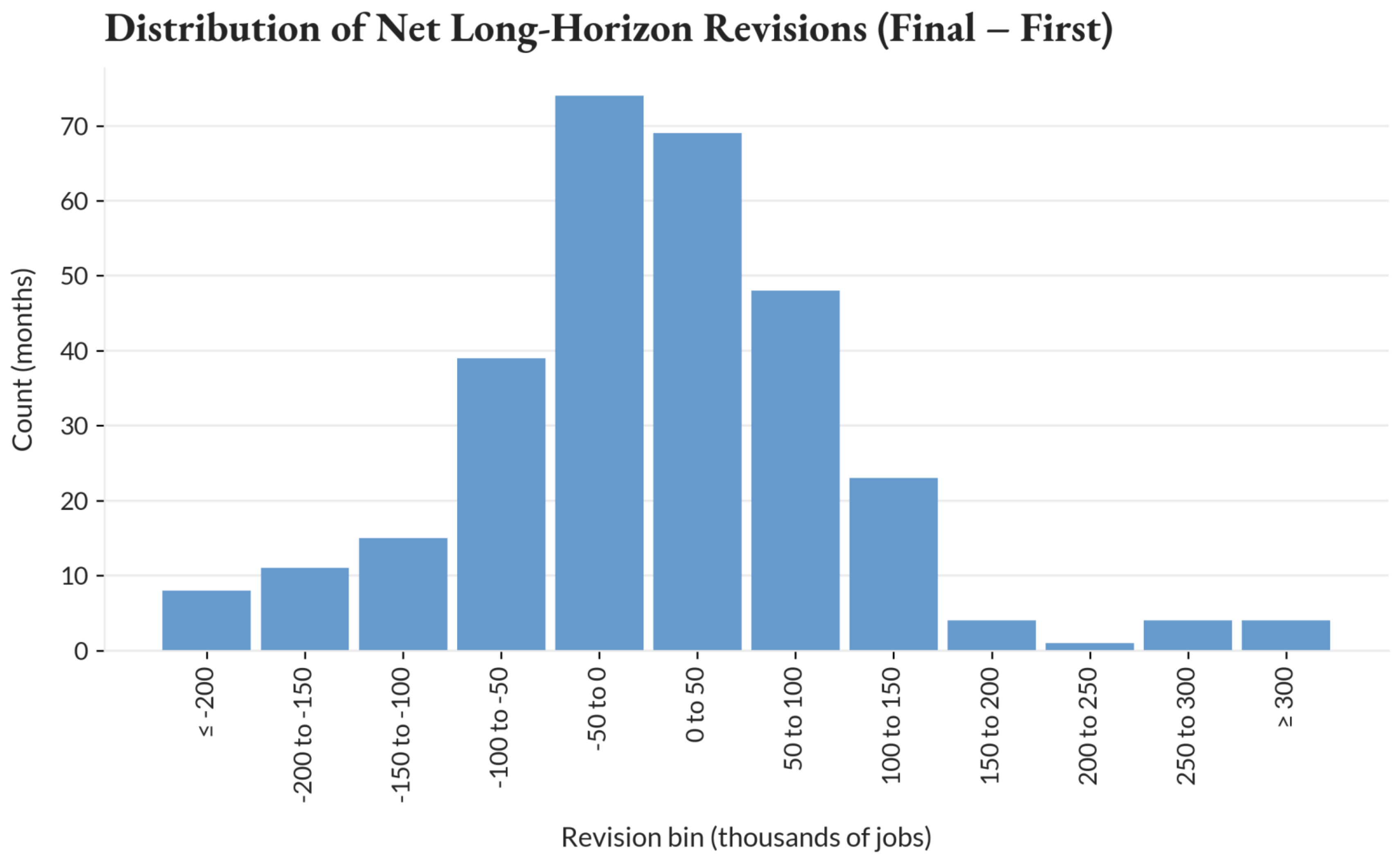
- Quantile regression forecasts revision distribution using lagged payroll signals and macro indicators available at release
- Calibration performance evaluated across economic regimes to test forecast accuracy during stable versus stress periods

02

Exploratory Data Analysis Evidence

How Big Are Revisions? What Effect Did COVID Have?

Small Median Masks Large Uncertainty



Insights

What the Distribution Shows

- Median near zero (3.5K non-COVID) but wide dispersion with standard deviation of 97K
- Heavy right tail extends beyond ±150K with visible extreme outliers
- Overflow bins capture revisions exceeding -200K and +300K

Key Statistical Properties

- Distribution shape rejects classical white noise measurement error
- Excess kurtosis of 2.6 confirms fatter tails than normal distribution
- Uncertainty magnitude varies widely month to month even outside COVID period

Exceedance Probabilities Show Tail Thickening

Summary Statistics for Long-Horizon Revisions (absolute, thousands of jobs)

Sample	Median	IQR	P95
Full Sample	53.0	71	209
Non-COVID	52.5	69	201
COVID-Only*	138.0	133	513

**COVID-only sample = March 2020 – December 2020 (n=10)*

Insights

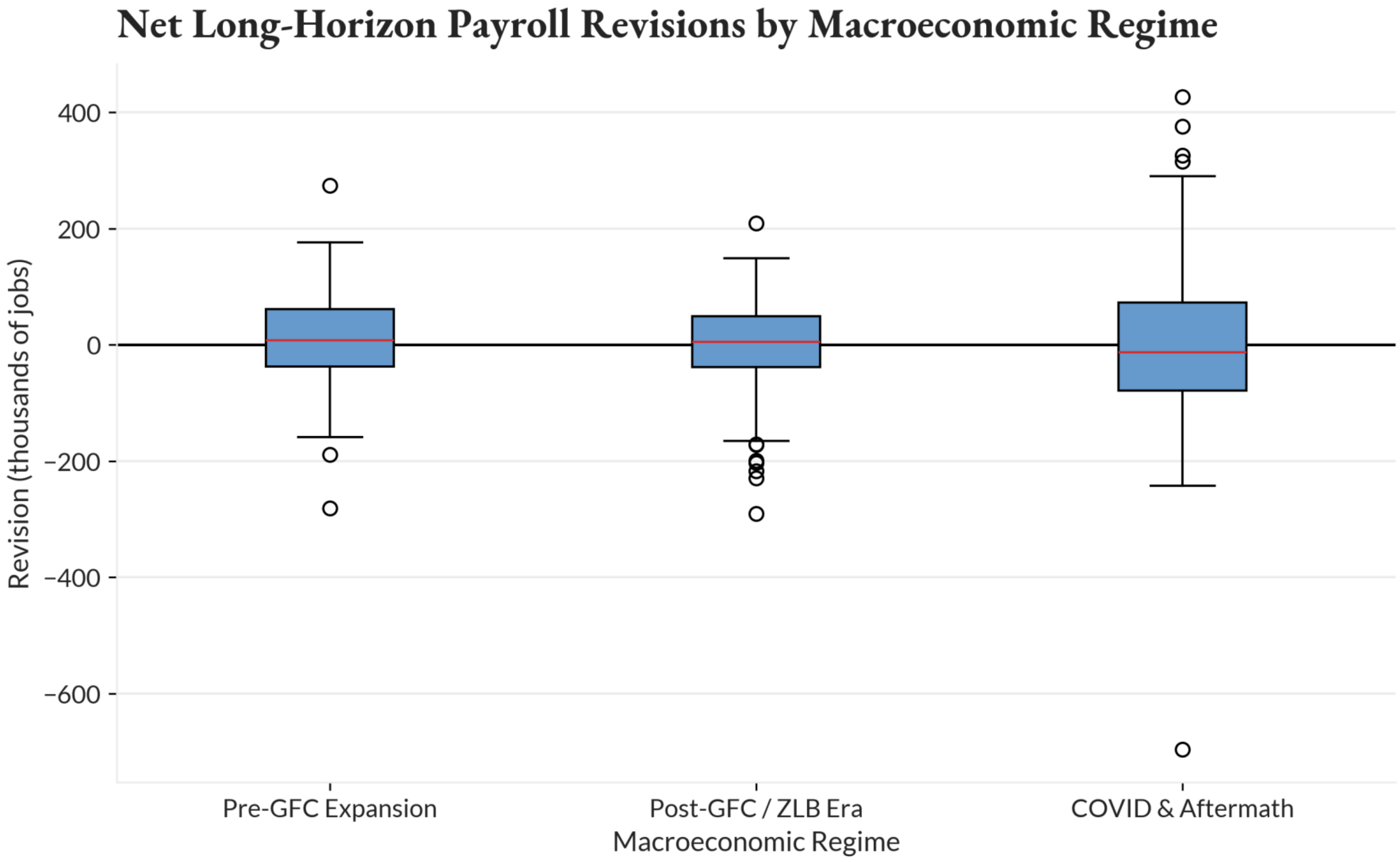
Baseline Uncertainty

- Median absolute revision of 52.5K even outside COVID
- IQR of 69K indicates substantial baseline dispersion
- 95th percentile of 201K shows large revisions occur regularly

COVID Amplification

- Median jumps from 52.5K to 138K (2.6× increase)
- P95 surges from 201K to 513K (2.6× increase in tail risk)
- IQR expands from 69K to 133K (1.9× increase in dispersion)

Revision Risk Varies by Macro Regime



Insights

Pre-COVID Stability

- Pre-GFC (2000-2007) and Post-GFC (2008-2019) show 20 years of mostly stable patterns
- Both baseline regimes have modest dispersion with IQR around 90K
- COVID & Aftermath (2020-2024) displays negative median and wider spread

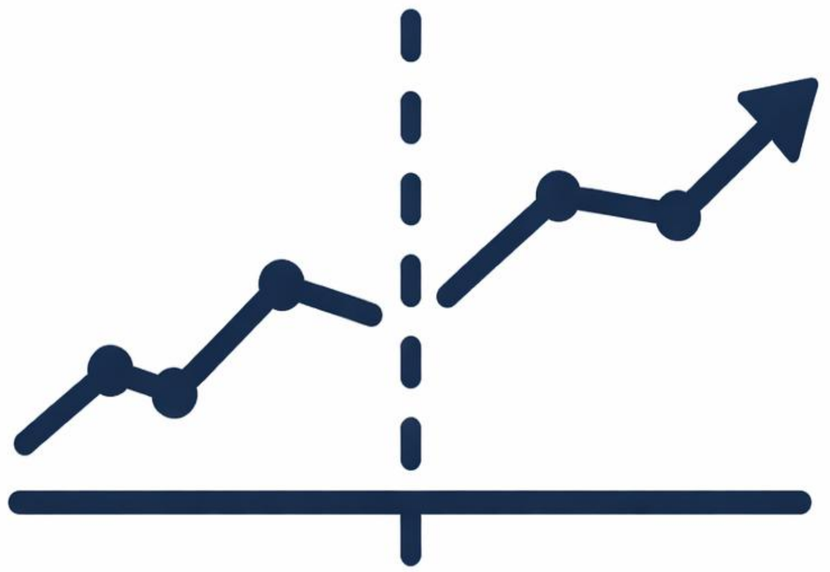
COVID Regime Shift

- IQR rises from 90K to 151K (1.7×) and persists across 60 months
- P95 doubles from 115K to 317K showing amplified tail risk
- Five-year duration indicates structural shift not temporary volatility spike

03

Interrupted Time Series Analysis

Testing for Structural Break at COVID Onset



Interrupted Time Series Model

An interrupted time series is estimated around the March 2020 COVID break:

$$\boldsymbol{Y_t = \beta_0 + \beta_1\, \mathrm{event\_time}_t + \beta_2\, \mathrm{post}_t + \beta_3(\mathrm{post}_t \times \mathrm{event\_time}_t) + \mathrm{Month}_t + \varepsilon_t}$$

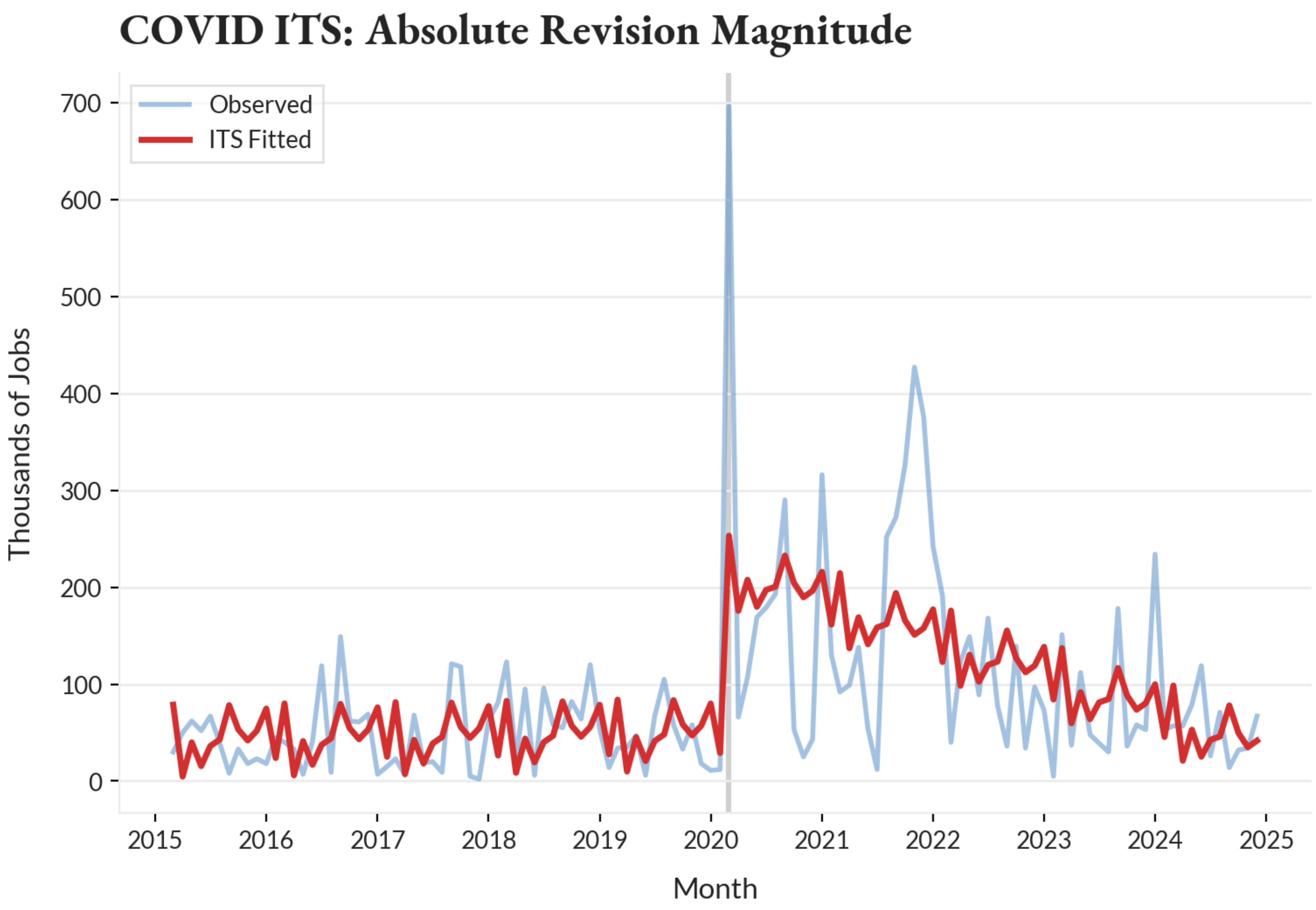
Where:

- Y_t is the absolute long-horizon revision magnitude or tail-risk probability
- $\mathrm{event\_time}_t$ is the month index relative to March 2020; post_t = 1 from March 2020 onward
- β_0 is the baseline level at event_time = 0 in the pre-COVID regime
- β_1 is the pre-COVID trend, the average change in Y_t per month before March 2020 (slope of the pre-period line).
- β_2 is the level shift at COVID onset; β_3 is the change in post-COVID slope
- Month_t is the monthly fixed effects; ε_t is the error term (HAC-robust inference)

The following identification assumption is made:

Without COVID, revision behavior would have continued along the pre-COVID linear trend with seasonal pattern.

How Did COVID Change Revision Magnitudes?



Insights

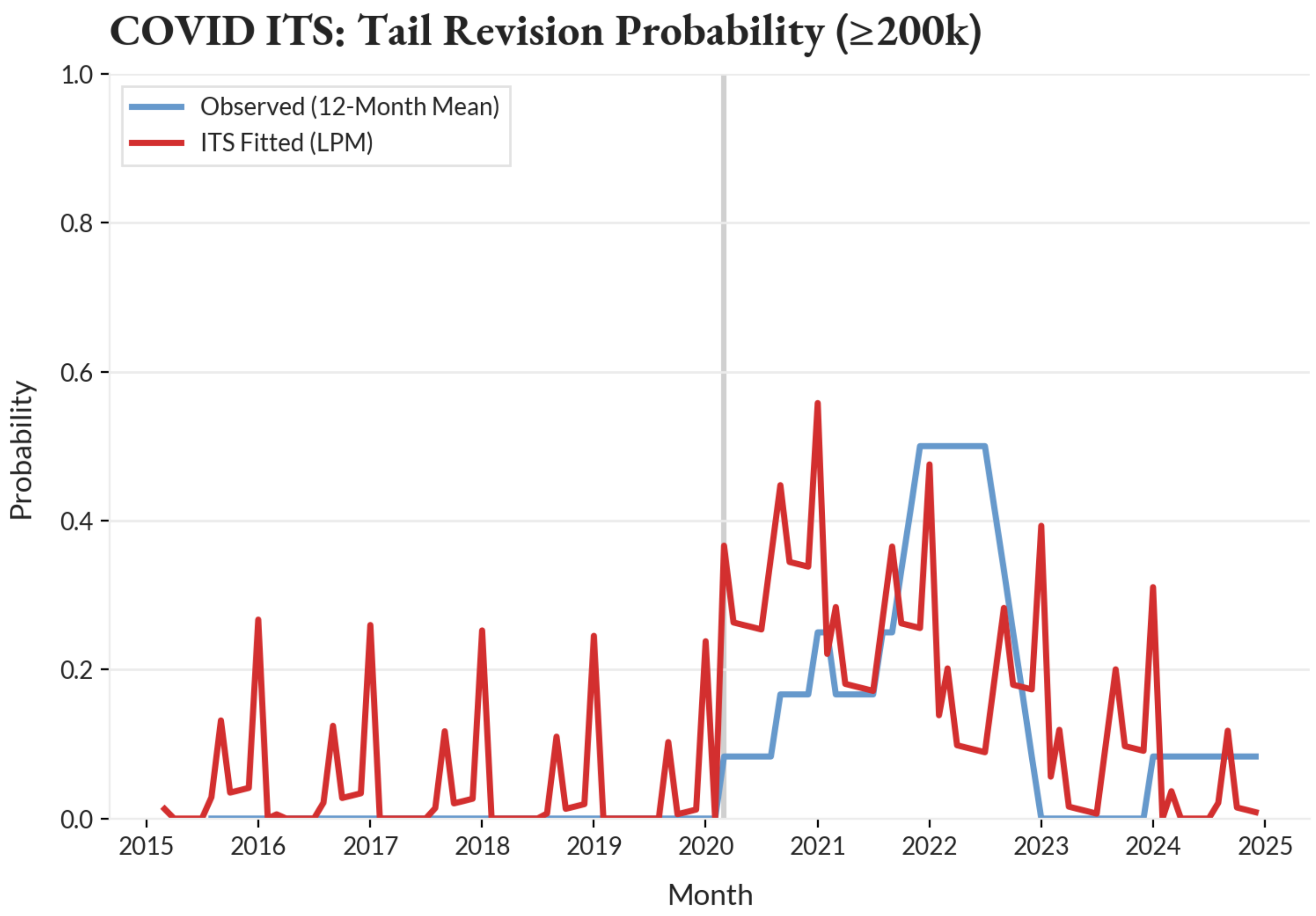
Level and Trend Shifts

- Pre-COVID trend remains approximately flat around the 50–100K range
- Level shift at COVID onset adds about 168K to revision magnitude
- Post-COVID slope declines by roughly 3.3K per month versus pre-COVID

Persistence Over Time

- Break at March 2020 produces an immediate jump in the fitted series
- Decay in revision magnitudes proceeds gradually over the 57 post-COVID months
- Elevated revision magnitudes persist for roughly four years before near-reversion

Tail Risk Has Not Returned to Baseline



Insights

The Decay Pattern

- Pre-COVID window shows zero 200K tail events (baseline 0%).
- Probability jumps about 39 percentage points at COVID onset ($p < 0.01$).
- Decay proceeds at roughly 0.63 percentage points per month, indicating slow reversion.

Why Incomplete Convergence Matters

- After 57 post-COVID months tail probability remains around 3.3 percentage points above baseline.
- Nearly five years of elevated tail risk is inconsistent with a purely transitory measurement shock.
- Extreme revisions stay structurally more likely even as average magnitudes partly normalize.

Placebo ITS Breaks for Absolute Net Revisions

Robustness Checks for COVID Break

Break date	Level shift (jobs, thousands)	p-value	Interpretation
March 2020 (COVID)*	+168	< 0.001	Real discontinuity
January 2016 (placebo)	−15	0.45	No meaningful effect
January 2017 (placebo)	−49	0.017	Modest negative shift
January 2018 (placebo)	+19	0.42	No meaningful effect

** COVID post window clipped to 57 months due to sample end*

Insights

Placebo Timing Evidence

- Placebo breaks (Jan 2016, 2017, 2018) yield small, imprecise shifts versus the large +168K jump at March 2020.
- Only COVID onset produces a statistically and economically significant discontinuity.
- This pattern confirms timing specificity rather than a spurious trend artifact.

Additional Robustness

- Pre-COVID trends are flat (event-time coefficients consistent with zero).
- Short-horizon revisions show no comparable March 2020 break (negative control).
- Results stable across 6, 12, and 18-month HAC lag specifications.

Interpreting the ITS Evidence: What It Shows—and What It Doesn't

What ITS Establishes

Evidence From Timing and Robustness Checks

Timing of the break

- Sharp March 2020 jump in revision magnitude and tail risk.
- No pre-COVID drift in event-time coefficients.

Persistence of effects

- Elevated revisions and 200K tail probability through 57 months.
- Tail risk not back to pre-COVID baseline.

Robustness

- Placebo breaks, pre-trends, and negative controls all align on a COVID-specific break.
- Results stable across HAC lag choices.

What ITS Cannot Establish

Limits of the Design

Mechanism

- Cannot separate response rates, birth-death modeling, and seasonal adjustment effects.
- Likely multiple operational channels.

Other changes in March 2020

- Institutional and policy shifts cannot be fully excluded.
- COVID timing makes them unlikely as sole drivers.

Functional form

- Segmented linear trend may miss nonlinear decay and plateaus.
- Alternative dynamics could change the path, not the existence, of the break.

04

Forecasting Revision Risk

Machine Learning vs. Fundamental Limits

Forecasting Revision Risk: Question & Approach

Why Forecast Revision Risk?

Decision Stakes and the Predictability Question

Can we forecast revision risk in real time?

- Using only data available at release
- Before revisions are observed

If predictable:

- Users adjust confidence dynamically
- Policy operates with calibrated risk awareness

If unpredictable:

- Static intervals mislead when risk spikes
- Policy decisions based on false precision

Methodological Approach

Forecasting Tail Risk With Rolling Out-Of-Sample Validation

Target: Tail risk quantiles

- 75th, 90th, 95th percentiles of revision magnitude
- Not average behavior

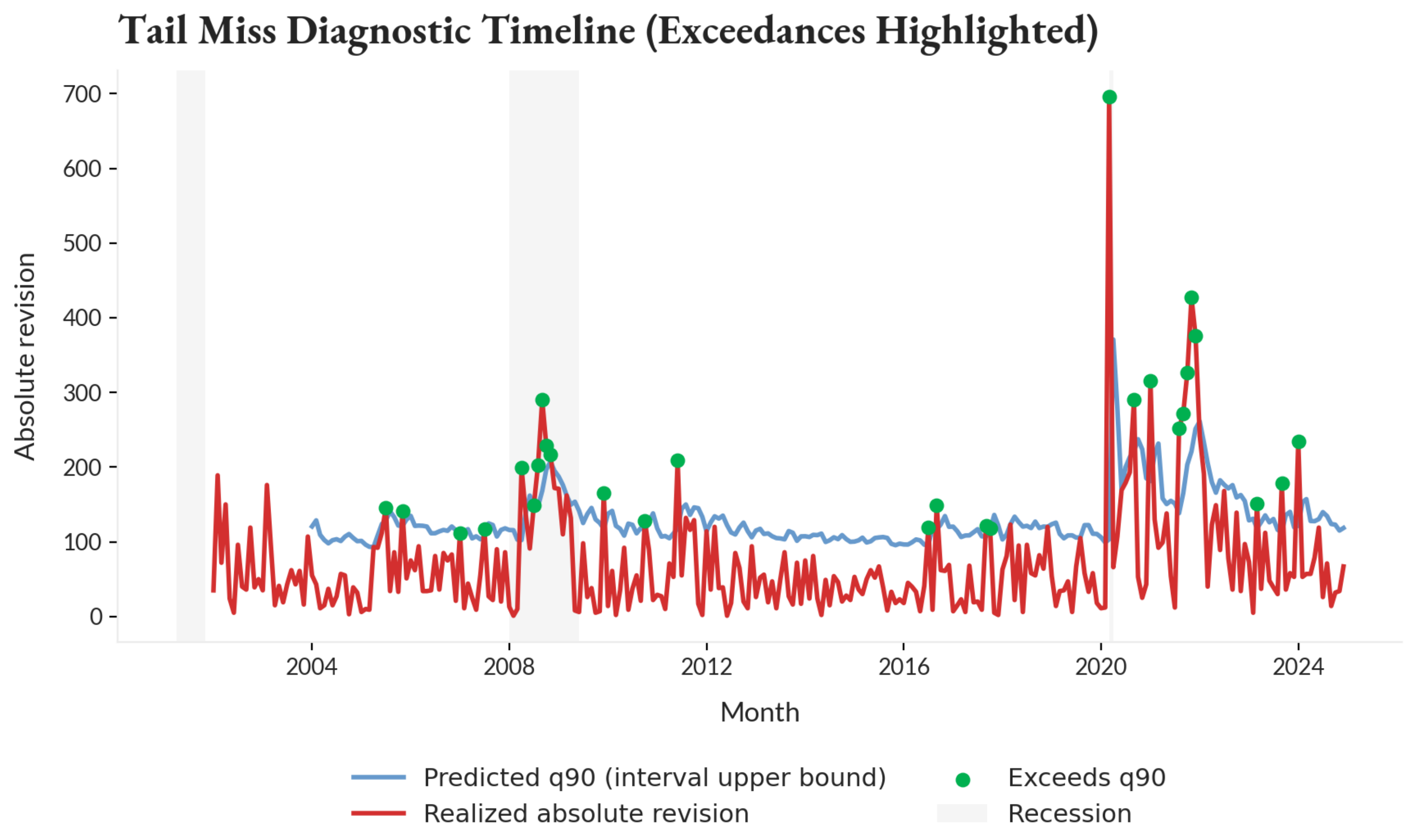
Features

- Lagged revisions, unemployment, recession indicator
- Rolling volatility, first estimate magnitude

Evaluation: Calibration

- Does predicted 90th percentile contain 90% of observations?
- Rolling window, strict out-of-sample (2010–2024)

Tail Risk Has Not Returned to Baseline



Insights

Tail Miss Patterns

- Pre-COVID tail events are rare and usually flagged by high model probabilities.
- COVID onset produces large tail misses (e.g., March 2020 and September 2020 exceed the 90th percentile).
- 2021–24 shows recurring tail misses, consistent with a higher-risk regime.

Why Incomplete Convergence Matters

- Pre-COVID–trained models cannot anticipate COVID-scale tail events.
- Post-COVID recalibration helps but still leaves surprise tails (e.g., January 2024).
- Regime shifts make some tail risk intrinsically hard to forecast from history alone.

Tail Risk Has Not Returned to Baseline

	Data/measurement issues	Model/inference issues
Root causes	**Survey Response Collapse** *Data Collection Infrastructure Under Stress* • COVID disrupted establishment reporting, sharply reducing CES sample coverage. • First releases increasingly rely on imputation rather than actual employer reports.	**Model Adjustment Breakdown** *Estimation Methods Trained on Stability* • Birth-death models and seasonal factors break when labor dynamics shift rapidly. • BLS operational constraints created persistent measurement volatility.
Consequences	**Forecasting Limits** *Historical Data Cannot Anticipate Disruption* • Pre-COVID training data embed stable regimes; operational shocks are out-of-sample. • No feature set can predict when survey infrastructure will break down.	**Ongoing Risk Elevation** *Implications for Real-Time Analysis* • Tail risk remains elevated four years post-COVID, signaling a regime shift. • Static confidence intervals assuming constant variance understate uncertainty.

05

From Evidence to Action

Implications for policy, markets, and real-time analysis

Policy Implications Across Three Stakeholders

Federal Reserve: Reliability varies across regimes, requiring conditional uncertainty

- Payroll employment is a key FOMC indicator, but revisions make its reliability regime-dependent.
- Consider supplementing with alternative indicators (jobless claims, ADP, JOLTS) when revision risk appears elevated.

Financial Markets: Standard pricing may understate revision risk during stress

- Options markets price in elevated volatility around payroll releases but may not fully account for regime-dependent revision risk.
- Tail risk models and volatility forecasts should incorporate evidence of time-varying uncertainty.

BLS: Make real-time uncertainty more visible and regime-sensitive

- Make existing sampling-error intervals more prominent alongside headline payroll changes.
- Develop regime-sensitive quality indicators (response rates, imputation shares) to flag when revision risk is unusually elevated.

Limitations and Future Research

Current Constraints

Scope and Identification Limits

- Cannot isolate specific COVID disruption channels (response rates, birth-death model, seasonal adjustment).
- Models are diagnostic, not operational; cannot predict future structural breaks.
- Limited to CES; CPS household survey divergence unexplored.

Methodological and Data Gaps

- Segmented linear trend may misspecify nonlinear decay dynamics.
- Cross-survey and cross-country comparisons not yet conducted.
- Data on response rates and imputation patterns not analyzed.

Future Research Directions

Real-Time Quality Enhancement

- Use real-time survey quality indicators (response rates, imputation shares, sample mix).
- Build Hybrid nowcasting combining private payroll data (ADP, Paychex) with official CES.
- Leverage early QCEW releases for real-time benchmark signals.

System-Level Comparative Analysis

- Analyze CES–CPS divergence during COVID.
- Compare systems cross-country (Europe, Canada, UK).
- Identify mechanisms with micro-data on response rates, imputation, and establishment churn.

Addressing these gaps would enable more reliable real-time inference and better-informed policy decisions.

Summary: Core Contributions and Key Findings

Four Key Findings

Heavy-Tailed Baseline
COVID Amplification
01.

Persistent Structural Break
57 Months Post-Onset
02.

Counter-Cyclical Risk
Turning Point Bias
03.

Calibration Breakdown
Fundamental Limits
04.

Research Contributions

Tail risk inherent to revision process

- COVID amplified existing risk nonlinearly, not creates new risk

Elevated tail risk remains

- Regime shift persists beyond typical shock dissipation window

Uncertainty peaks when data is most critical

- Measurement reliability lowest at business cycle inflection points

Real-time forecasting fails under stress

- This reflects underlying measurement instability, not a model deficiency

Measurement Uncertainty Matters—Especially When It Matters Most

Thank you! Full code & analysis available on GitHub.

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